Project Proposal Form
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Project Title: AutoRestTest: An AI-Powered Platform for Automated REST API Testing

Motivation: As modern web services heavily rely on REST APIs, ensuring API correctness is crucial. When APIs grow in size and complexity, testing them becomes increasingly difficult due to large input spaces, interdependent operations and the need for valid request sequences. Traditional automated testing tools often fail to effectively explore these dependencies, leading to low coverage, missing edge cases and less effectiveness in fault detection. Manual testing is also time-consuming, repetitive, error-prone and not scalable for continuous development workflows. Recent research, such as AutoRestTest [1], demonstrates that dependency-aware, LLM-powered, reinforcement learning–based test generation can significantly improve fault detection and coverage in REST APIs. At the same time, development teams require a centralized, collaborative platform where test suites, execution results, reports and project data can be stored and reused over time. This project addresses these challenges by proposing an AI-powered collaborative SaaS platform for automated REST API testing, combining an intelligent testing engine inspired by the AutoRestTest approach with a user-friendly system for managing projects, test suites, and shared testing workflows.

Proposed Solution: The proposed system will be built using an intelligent approach that integrates Multi-Agent Reinforcement Learning (MARL) with a Semantic Property Dependency Graph (SPDG) and Large Language Models (LLMs). The system will take an OpenAPI specification file or automatically generate one from the provided codebase using AI-assisted analysis and then analyze the API structure and construct a dependency graph representing potential dependencies between operations. The system will then utilize multiple intelligent agents responsible for selecting API operations, choosing parameter combinations, generating parameter values and managing dependencies between requests. This approach utilizes dependency-aware request generation, adaptive learning from previous responses and AI-assisted input generation to improve test effectiveness and coverage. The system will also apply request mutations to identify unexpected behaviors and potential server-side failures. Finally, the platform will generate a detailed report containing tested operations, status-code distribution and discovered server errors for future use.

Project Features: The platform will provide a collaborative and user-friendly environment for automated REST API testing. Users will be able to create accounts, manage multiple projects, upload OpenAPI specification files or automatically generate OpenAPI specifications using AI-assisted analysis and organize automated test suites through a centralized web dashboard. The system will support intelligent one-click API testing, automated workflow discovery, execution history tracking, and detailed testing reports containing status-code distributions and detected server errors. In addition, users will be able to save previous test results for future analysis, compare repeated executions, and collaborate with multiple team members within the same project. By automating complex API testing workflows and reducing manual effort, the platform aims to improve testing efficiency, increase API reliability, and simplify REST API quality assurance for development teams.

Languages & Technologies:

- Language: Python, Typescript
- Frontend: NextJS, TailwindCSS
- Backend: NestJS
- Database: PostgreSQL, Prisma ORM
- LLM Provider: OpenRouter
- Version control: Git, Github
- IDE & other softwares: VS Code, Postman

References:

[1] Tyler Stennett, Myeongsoo Kim, Saurabh Sinha, and Alessandro Orso,
A Multi-Agent Approach for REST API Testing with Semantic Graphs and LLM-Driven Inputs, in Proceedings of 47th International Conference on Software Engineering (ICSE 2025)
Paper: https://arxiv.org/abs/2411.07098v2

Supervisor's Name: Toukir Ahammed

Signature of the supervisor:

Date: 17th May, 2026

Proposal Presentation Feedback:

Midterm Presentation Feedback: